

PsExec Remote Execution Detection and Investigation using Sysmon and WazuhUntitled

Executive Summary

This investigation simulated PsExec-based remote execution from a Kali Linux system against a monitored Windows endpoint. The objective was to analyze how Windows Security Logs, Sysmon, Windows Defender, and Wazuh detect remote authentication, service creation, payload deployment, named pipe communication, remote code execution, and post-compromise discovery activity.

Initial execution attempts were detected and blocked by Microsoft Defender. Telemetry analysis was subsequently performed using a successful execution scenario to capture the complete attack lifecycle and validate detection visibility across multiple logging sources.

The investigation demonstrated a complete attack chain consisting of:

- Remote NTLM authentication
- Administrative privilege assignment
- Payload transfer through ADMIN\$
- Remote service creation
- Service-based execution
- Named pipe communication
- Remote shell establishment
- Discovery command execution
- Wazuh alert generation
- Session termination

Objective

The objective of this investigation was to identify and correlate security telemetry generated during PsExec-based remote execution and lateral movement activity using:

- Windows Security Events
- Windows System Events
- Sysmon
- Windows Defender
- Wazuh SIEM

MITRE ATT&CK Mapping

Tactic	Technique	ID
Lateral Movement	SMB/Windows Admin Shares	T1021.002
Lateral Movement	Remote Services	T1021
Execution	Service Execution	T1569.002
Persistence	Windows Service	T1543.003
Defense Evasion	Valid Accounts	T1078
Discovery	System Owner/User Discovery	T1033
Discovery	System Information Discovery	T1082
Discovery	Process Discovery	T1057
Discovery	System Network Configuration Discovery	T1016
Lateral Movement	Lateral Tool Transfer	T1570

Attack Simulation

The following command was executed from the Kali Linux system:

```
impacket-psexec socadmin: 'password'@192.168.56.106
```

Observed activity:

```
Found writable share ADMIN$  
Uploading file ahsjitDb.exe  
Creating service lnsc  
Starting service lnsc
```

The payload was uploaded through the ADMIN\$ share and executed as a temporary Windows service running under LocalSystem privileges.

```
(kali㉿kali)-[~]  
$ impacket-psexec socadmin:'kali'@192.168.56.106  
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies  
  
[*] Requesting shares on 192.168.56.106.....  
[*] Found writable share ADMIN$  
[*] Uploading file ahsjitDb.exe  
[*] Opening SVCManager on 192.168.56.106.....  
[*] Creating service lnsc on 192.168.56.106.....  
[*] Starting service lnsc.....  
[!] Press help for extra shell commands  
Microsoft Windows [Version 10.0.26200.8037]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Windows\System32> whoami
```

Attack Chain Overview

```
NTLM Authentication  
    ↓  
Administrative Access  
    ↓  
Payload Upload  
    ↓  
Service Creation  
    ↓  
Service Execution  
    ↓  
Named Pipe Creation  
    ↓  
Named Pipe Communication  
    ↓  
Remote Shell Established  
    ↓  
Discovery Commands Executed  
    ↓  
Session Termination
```

Initial Prevention Attempt

Prior to successful execution, Microsoft Defender identified the uploaded PsExec service executable as:

```
VirTool:Win32/RemoteExec!pz
```

Observed Events:

Event ID	Description
1116	Malware Detection
1117	Malware Quarantine

Detected File:

```
C:\Windows\ahsjitDb.exe
```

Detected Service:

```
lnsc
```

This demonstrates that modern endpoint protection solutions can identify PsExec-style execution frameworks before execution is completed.

No further analysis was performed on these events because the primary investigation focused on the successful attack lifecycle.

Remote Authentication Analysis

Windows Security Event ID 4624

Findings

Field	Value
Event ID	4624
Username	socadmin
Source IP	192.168.56.107
Logon Type	3
Authentication Package	NTLM
NTLM Version	NTLM V2
Elevated Token	Yes

Analysis

Windows recorded a successful remote network authentication from the Kali Linux system.

Important Indicators:

Logon Type: 3
Authentication Package: NTLM
NTLM Version: NTLM V2
Source IP: 192.168.56.107
Account: socadmin
Elevated Token: Yes

This event established the initial access phase required for remote service execution.

Security

Number of events: 181 (!) New events available

Keywords	Date and Time	Source	Event ID	Task Category
Audit Suc...	6/1/2026 11:16:47 AM	Microsoft Wi...	4624	Logon

Event 4624, Microsoft Windows security auditing.

General

Details

An account was successfully logged on.

Subject:

Security ID:

Account Name:

Account Domain:

Logon ID:

NULL SID

-

-

0x0

Logon Information:

Logon Type:

Restricted Admin Mode:

Remote Credential Guard:

Virtual Account:

Elevated Token:

3

-

-

No

Yes

Impersonation Level:

Impersonation

New Logon:

Security ID:

Account Name:

Account Domain:

Logon ID:

Linked Logon ID:

Network Account Name:

Network Account Domain:

Logon GUID:

DESKTOP-HDF09SQ\socadmin

socadmin

DESKTOP-HDF09SQ

0x1174957

0x0

-

-

{00000000-0000-0000-0000-000000000000}

Network Information:

Workstation Name:

Source Network Address:

Source Port:

-

192.168.56.107

56450

Detailed Authentication Information:

Logon Process:

Authentication Package:

Transited Services:

Package Name (NTLM only):

Key Length:

NtLmSsp

NTLM

-

NTLM V2

128

This event is generated when a logon session is created. It is generated on the computer that was accessed.

The subject fields indicate the account on the local system which requested the logon. This is most commonly a service such as the Server service, or a local process such as Winlogon.exe or Services.exe.

The logon type field indicates the kind of logon that occurred. The most common types are 2 (interactive) and 3 (network).

The New Logon fields indicate the account for whom the new logon was created, i.e. the account that was logged on.

The network fields indicate where a remote logon request originated. Workstation name is not always available and may be left blank in some cases.

The impersonation level field indicates the extent to which a process in the logon session can

Log Name:

Source:

Event ID:

Level:

User:

OpCode:

Security

Microsoft Windows security

4624

Information

N/A

Info

Logged:

Task Category:

Keywords:

Computer:

6/1/2026 11:16:47 AM

Logon

Audit Success

DESKTOP-HDF09SQ

Activate Wi

Administrative Privilege Assignment

Windows Security Event ID 4672

Findings

Administrative privileges were assigned immediately after authentication.

Observed Privileges:

```
SeDebugPrivilege
SeBackupPrivilege
SeRestorePrivilege
SeTakeOwnershipPrivilege
SeLoadDriverPrivilege
SeImpersonatePrivilege
```

Analysis

This event confirmed that the authenticated account possessed sufficient privileges to perform service creation and remote execution operations.

Security Number of events: 181 (!) New events available

Keywords	Date and Time	Source	Event ID	Task Category
Audit Suc...	6/1/2026 11:16:47 AM	Microsoft Wi...	4672	Special Logon

Event 4672, Microsoft Windows security auditing.

General Details

Special privileges assigned to new logon.

Subject:

Security ID: DESKTOP-HDF09SQ\socadmin
Account Name: socadmin
Account Domain: DESKTOP-HDF09SQ
Logon ID: 0x1174957

Privileges:

SeSecurityPrivilege
SeBackupPrivilege
SeRestorePrivilege
SeTakeOwnershipPrivilege
SeDebugPrivilege
SeSystemEnvironmentPrivilege
SeLoadDriverPrivilege
SeImpersonatePrivilege
SeDelegateSessionUserImpersonatePrivilege

Log Name: Security
Source: Microsoft Windows security Logged: 6/1/2026 11:16:47 AM
Event ID: 4672 Task Category: Special Logon
Level: Information Keywords: Audit Success
User: N/A Computer: DESKTOP-HDF09SQ
OpCode: Info

Payload Transfer Analysis

Sysmon Event ID 11 – File Creation

Findings

Field	Value
Event ID	11
Process	System
File Created	C:\Windows\ahsjitDb.exe
User	NT AUTHORITY\SYSTEM

Analysis

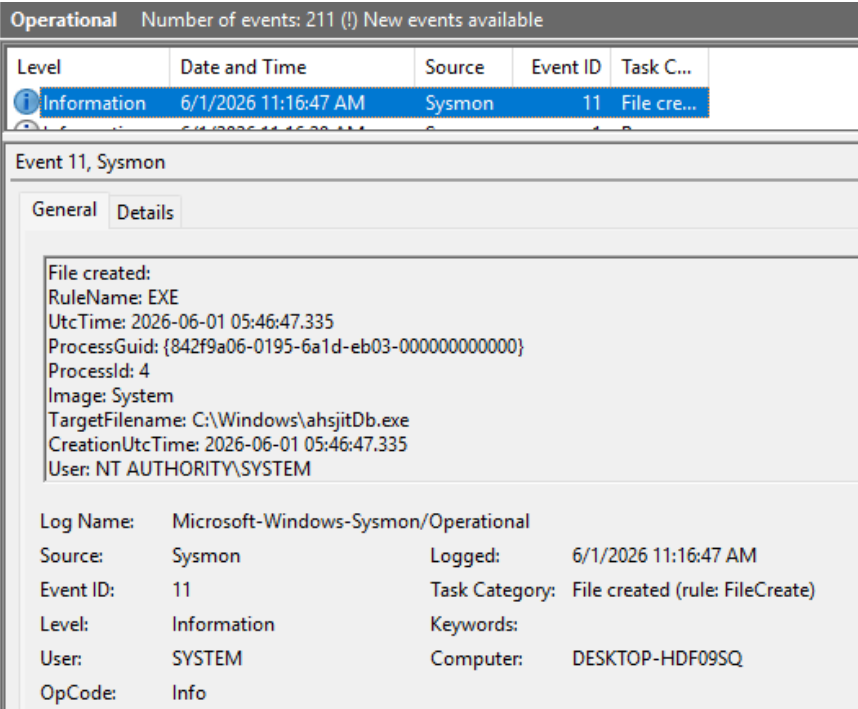
Sysmon detected creation of the uploaded PsExec service executable.

```
C:\Windows\ahsjitDb.exe
```

This represents lateral tool transfer through the ADMIN\$ administrative share.
MITRE:

T1570

Lateral Tool Transfer



Payload Execution Analysis

Sysmon Event ID 1 – Process Creation

Findings

Field	Value
Event ID	1
Process	C:\Windows\ahsjitDb.exe
Parent Process	services.exe
User	NT AUTHORITY\SYSTEM

Analysis

Sysmon Event ID 1 confirmed execution of the uploaded service executable. This event provides direct evidence that the transferred payload successfully executed under LocalSystem privileges.

Indicators:

Parent Process: services.exe
Child Process: ahsjitDb.exe
User: SYSTEM

This event marks the transition from payload staging to active remote code execution.

Operational

Number of events: 211 (!) New events available

Level	Date and Time	Source	Event ID	Task C...
Information	6/1/2026 11:16:47 AM	Sysmon	1	Proces...

Event 1, Sysmon

General

Details

Process Create:

RuleName: -

UtcTime: 2026-06-01 05:46:47.414

ProcessGuid: {842f9a06-1cc7-6a1d-3403-000000002700}

ProcessId: 3440

Image: C:\Windows\ahsjitDb.exe

FileVersion: -

Description: -

Product: -

Company: -

OriginalFileName: -

CommandLine: C:\WINDOWS\ahsjitDb.exe

CurrentDirectory: C:\WINDOWS\system32\

User: NT AUTHORITY\SYSTEM

LogonGuid: {842f9a06-019c-6a1d-e703-000000000000}

LogonId: 0x3E7

TerminalSessionId: 0

IntegrityLevel: System

Hashes: MD5=6983F7001DE10F4D19FC2D794C3EB534,SHA256=3C2FE308C0A563E06263BBACF793BBE9B2259D795FCC36B953793A7E499E7F71,IMPHASH=4749670AC3D28D6761142B0DCB4F5076

ParentProcessGuid: {842f9a06-019c-6a1d-0b00-000000002700}

ParentProcessId: 852

ParentImage: C:\Windows\System32\services.exe

ParentCommandLine: C:\WINDOWS\system32\services.exe

ParentUser: NT AUTHORITY\SYSTEM

Log Name: Microsoft-Windows-Sysmon/Operational

Source: Sysmon

Event ID: 1

Level: Information

User: SYSTEM

OpCode: Info

Logged: 6/1/2026 11:16:47 AM

Task Category: Process Create (rule: ProcessCreate)

Keywords:

Computer: DESKTOP-HDF09SQ

Activa

Service Creation Analysis

Windows Event ID 7045

Findings

Field	Value
Event ID	7045
Service Name	Insc
Service Binary	%systemroot%\ahsjitDb.exe
Account	LocalSystem

Analysis

Windows recorded installation of a new service.

Service Name: Insc
Service Binary: %systemroot%\ahsjitDb.exe

This event is one of the strongest indicators of PsExec activity because PsExec relies on temporary service creation to launch remote commands.

MITRE:

T1569.002 : Service Execution

System Number of events: 33

Level	Date and Time	Source	Event ID	Task Catego...
Information	6/1/2026 11:16:47 AM	Service Con...	7045	None

Event 7045, Service Control Manager

General

Details

A service was installed in the system.

Service Name: Insc
Service File Name: %systemroot%\ahsjitDb.exe
Service Type: user mode service
Service Start Type: demand start
Service Account: LocalSystem

Log Name: System
Source: Service Control Manager Logged: 6/1/2026 11:16:47 AM
Event ID: 7045 Task Category: None
Level: Information Keywords: Classic
User: DESKTOP-HDF09SQ\socadm Computer: DESKTOP-HDF09SQ
OpCode: Info

Registry-Based Service Creation

Sysmon Event ID 13

Findings

Registry Keys Created:

```
HKLM\System\CurrentControlSet\Services\lnsc\Start
HKLM\System\CurrentControlSet\Services\lnsc\ImagePath
```

Binary:

```
%systemroot%\ahsjitDb.exe
```

Analysis

Sysmon detected service registration activity within the Windows Registry.
This confirms:

- Service installation
- Service configuration
- Service execution path registration

MITRE:

T1543.003 : Windows Service

Operational

Number of events: 233

Level	Date and Time	Source	Event ID	Task C...
 Information	6/1/2026 11:16:47 AM	Sysmon	13	Registr...

Event 13, Sysmon

General

Details

Registry value set:
RuleName: T1031,T1050
EventType: SetValue
UtcTime: 2026-06-01 05:46:47.363
ProcessGuid: {842f9a06-019c-6a1d-0b00-000000002700}
ProcessId: 852
Image: C:\WINDOWS\system32\services.exe
TargetObject: HKLM\System\CurrentControlSet\Services\lnsc\Start
Details: DWORD (0x00000003)
User: NT AUTHORITY\SYSTEM

Log Name:

Microsoft-Windows-Sysmon/Operational

Source:

Sysmon

Logged:

6/1/2026 11:16:47 AM

Event ID:

13

Task Category:

Registry value set (rule: RegistryEvent)

Level:

Information

Keywords:

User:

SYSTEM

Computer:

DESKTOP-HDF09SQ

OpCode:

Info

Operational
Number of events: 233 (!) New events available

Level	Date and Time	Source	Event ID	Task C...
Information	6/1/2026 11:16:47 AM	Sysmon	13	Registr...

Event 13, Sysmon

General
Details

Registry value set:
RuleName: T1031,T1050
EventType: SetValue
UtcTime: 2026-06-01 05:46:47.363
ProcessGuid: {842f9a06-019c-6a1d-0b00-000000002700}
ProcessId: 852
Image: C:\WINDOWS\system32\services.exe
TargetObject: HKLM\System\CurrentControlSet\Services\Insc\ImagePath
Details: %%systemroot%%\ahsjitDb.exe
User: NT AUTHORITY\SYSTEM

Log Name: Microsoft-Windows-Sysmon/Operational
Source: Sysmon
Event ID: 13
Level: Information
User: SYSTEM
OpCode: Info

Logged: 6/1/2026 11:16:47 AM
Task Category: Registry value set (rule: RegistryEvent)
Keywords:
Computer: DESKTOP-HDF09SQ

Pipe Communication Analysis

Sysmon Event ID 17 – Pipe Created

Findings

Field	Value
Event ID	17
Process	C:\Windows\ahsjitDb.exe
Pipe Name	\RemCom_communicaton

Analysis

The PsExec service executable created the following named pipe:

```
\RemCom_communicaton
```

This is a strong indicator of Impacket PsExec activity because the RemCom framework is used to establish communication between the attacker and the remote service.

Operational Number of events: 233 (!) New events available

Level	Date and Time	Source	Event ID	Task C...
Information	6/1/2026 11:16:47 AM	Sysmon	17	Pipe Cr...

Event 17, Sysmon

GeneralDetails

Pipe Created:
RuleName: -
EventType: CreatePipe
UtcTime: 2026-06-01 05:46:47.435
ProcessGuid: {842f9a06-1cc7-6a1d-3403-000000002700}
ProcessId: 3440
PipeName: \RemCom_communicaton
Image: C:\WINDOWS\ahsjitDb.exe
User: NT AUTHORITY\SYSTEM

Log Name: Microsoft-Windows-Sysmon/Operational
Source: Sysmon Logged: 6/1/2026 11:16:47 AM
Event ID: 17 Task Category: Pipe Created (rule: PipeEvent)
Level: Information Keywords:
User: SYSTEM Computer: DESKTOP-HDF09SQ
OpCode: Info

Sysmon Event ID 18 – Pipe Connected Findings

Field	Value
Event ID	18
Process	System
Pipe Name	\RemCom_communicaton

Analysis

Sysmon recorded a successful connection to the named pipe.
Observed sequence:

```
CreatePipe
      ↓
ConnectPipe
```

This confirms active communication between the PsExec service and the Windows operating system.

Operational				
Number of events: 233 (!) New events available				
Level	Date and Time	Source	Event ID	Task C...
Information	6/1/2026 11:16:47 AM	Sysmon	18	Pipe C...

Event 18, Sysmon

General

Details

Pipe Connected:
RuleName: -
EventType: ConnectPipe
UtcTime: 2026-06-01 05:46:47.446
ProcessGuid: {842f9a06-0195-6a1d-eb03-000000000000}
ProcessId: 4
PipeName: \RemCom_communicaton
Image: System
User: NT AUTHORITY\SYSTEM

Log Name: Microsoft-Windows-Sysmon/Operational

Source:	Sysmon	Logged:	6/1/2026 11:16:47 AM
Event ID:	18	Task Category:	Pipe Connected (rule: PipeEvent)
Level:	Information	Keywords:	
User:	SYSTEM	Computer:	DESKTOP-HDF09SQ
OpCode:	Info		

Additional Pipe Evidence

Additional pipes identified:

```
\RemCom_stderrwakm51514 ; \RemCom_stdinwakm51514
```

Purpose:

Pipe	Function
stdin	Receive attacker commands
stdout	Return command output
stderr	Return command errors

Operational Number of events: 233 (!) New events available				
Level	Date and Time	Source	Event ID	Task C...
 Information	6/1/2026 11:16:47 AM	Sysmon	17	Pipe Cr...

Event 17, Sysmon

General

Details

Pipe Created:
RuleName: -
EventType: CreatePipe
UtcTime: 2026-06-01 05:46:47.449
ProcessGuid: {842f9a06-1cc7-6a1d-3403-000000002700}
ProcessId: 3440
PipeName: \RemCom_stdinwakm51514
Image: C:\WINDOWS\ahsjitDb.exe
User: NT AUTHORITY\SYSTEM

Log Name:

Microsoft-Windows-Sysmon/Operational

Source:

Sysmon

Logged:

6/1/2026 11:16:47 AM

Event ID:

17

Task Category:

Pipe Created (rule: PipeEvent)

Level:

Information

Keywords:

User:

SYSTEM

Computer:

DESKTOP-HDF09SQ

OpCode:

Info

Operational

Number of events: 233 (!) New events available

Level	Date and Time	Source	Event ID	Task C...
 Information	6/1/2026 11:16:47 AM	Sysmon	18	Pipe C...

Event 18, Sysmon

General

Details

Pipe Connected:
RuleName: -
EventType: ConnectPipe
UtcTime: 2026-06-01 05:46:47.467
ProcessGuid: {842f9a06-0195-6a1d-eb03-000000000000}
ProcessId: 4
PipeName: \RemCom_stdinwakm51514
Image: System
User: NT AUTHORITY\SYSTEM

Log Name: Microsoft-Windows-Sysmon/Operational

Source: Sysmon

Logged: 6/1/2026 11:16:47 AM

Event ID: 18

Task Category: Pipe Connected (rule: PipeEvent)

Level: Information

Keywords:

User: SYSTEM

Computer: DESKTOP-HDF09SQ

OpCode: Info

This confirms establishment of a fully interactive remote shell.

Remote Shell Activity

Following successful service execution, a SYSTEM-level shell was established.Commands executed :

```
whoami
hostname
ipconfig /all
netuser
tasklist
sc query
```

MITRE Mapping

Command	Technique
whoami	T1033
hostname	T1082
ipconfig	T1016
tasklist	T1057

These commands represent typical post-compromise discovery activity.

```
C:\Windows\System32> whoami
nt authority\system

C:\Windows\System32> hostname
DESKTOP-HDF09SQ
```

```
C:\Windows\System32> ipconfig /all
```

Windows IP Configuration

```
Host Name . . . . . : DESKTOP-HDF09SQ
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
```

Ethernet adapter Ethernet:

```
Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
Physical Address. . . . . : 08-00-27-EC-4A-15
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : fd17:625c:f037:2:fad8:9cb8:ccda:c957(Preferred)
Temporary IPv6 Address. . . . . : fd17:625c:f037:2:b8fb:f80a:b464:e4a(Preferred)
Link-local IPv6 Address . . . . . : fe80::950d:2704:ad74:aaf3(Preferred)
IPv4 Address. . . . . : 10.0.2.15(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Monday, June 1, 2026 9:20:55 AM
Lease Expires . . . . . : Tuesday, June 2, 2026 9:20:43 AM
Default Gateway . . . . . : fe80::2%3
                          10.0.2.2
DHCP Server . . . . . : 10.0.2.2
DHCPv6 IAID . . . . . : 235405351
DHCPv6 Client DUID. . . . . : 00-01-01-00-31-9F-B3-9D-08-00-27-EC-4A-15
DNS Servers . . . . . : 192.168.1.1
                          114.114.114.114
                          8.8.8.8
NetBIOS over Tcpip. . . . . : Enabled
```

Ethernet adapter Ethernet 2:

```
Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter #2
Physical Address. . . . . : 08-00-27-38-67-13
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::94e1:adfd:4ac6:c8bd%6(Preferred)
IPv4 Address. . . . . : 192.168.56.106(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Monday, June 1, 2026 10:41:11 AM
Lease Expires . . . . . : Monday, June 1, 2026 11:26:19 AM
Default Gateway . . . . . :
DHCP Server . . . . . : 192.168.56.100
DHCPv6 IAID . . . . . : 302514215
DHCPv6 Client DUID. . . . . : 00-01-01-00-31-9F-B3-9D-08-00-27-EC-4A-15
NetBIOS over Tcpip. . . . . : Enabled
```

```
C:\Windows\System32> whoami /priv
```

```
C:\Windows\System32> net user
```

User accounts for \\

```
Administrator      DefaultAccount      Guest
socadmin            WDAGUtilityAccount
```

The command completed with one or more errors.

```
C:\Windows\System32> net localgroup administrators
```

```
Alias name      administrators
Comment        Administrators have complete and unrestricted access to the com
```

Members

```
Administrator
socadmin
The command completed successfully.
```

```
C:\Windows\System32> tasklist
```

Image Name	PID	Session Name	Session#	Mem Usage
System Idle Process	0	Services	0	8 K
System	4	Services	0	128 K
Registry	116	Services	0	27,648 K
smss.exe	468	Services	0	232 K
csrss.exe	636	Services	0	2,152 K
wininit.exe	708	Services	0	432 K
csrss.exe	716	Console	1	1,988 K
winlogon.exe	780	Console	1	3,316 K
services.exe	852	Services	0	8,100 K
lsass.exe	876	Services	0	13,396 K
svchost.exe	1004	Services	0	23,968 K
fontdrvhost.exe	8	Console	1	760 K
fontdrvhost.exe	552	Services	0	164 K
svchost.exe	600	Services	0	12,304 K
svchost.exe	848	Services	0	3,848 K
dwm.exe	1104	Console	1	66,412 K
svchost.exe	1192	Services	0	11,428 K
svchost.exe	1212	Services	0	3,128 K

```
C:\Windows\System32> sc query

SERVICE_NAME: AppIDSvc
DISPLAY_NAME: Application Identity
        TYPE               : 30  WIN32
        STATE                : 4  RUNNING
                        (STOPPABLE, NOT_PAUSABLE, ACCEPTS_SHUTDOWN)
        WIN32_EXIT_CODE       : 0  (0x0)
        SERVICE_EXIT_CODE    : 0  (0x0)
        CHECKPOINT           : 0x0
        WAIT_HINT            : 0x0

SERVICE_NAME: Appinfo
DISPLAY_NAME: Application Information
        TYPE               : 30  WIN32
        STATE                : 4  RUNNING
                        (STOPPABLE, NOT_PAUSABLE, IGNORES_SHUTDOWN)
        WIN32_EXIT_CODE       : 0  (0x0)
        SERVICE_EXIT_CODE    : 0  (0x0)
        CHECKPOINT           : 0x0
        WAIT_HINT            : 0x0

SERVICE_NAME: AudioEndpointBuilder
DISPLAY_NAME: Windows Audio Endpoint Builder
        TYPE               : 30  WIN32
        STATE                : 4  RUNNING
                        (STOPPABLE, NOT_PAUSABLE, IGNORES_SHUTDOWN)
        WIN32_EXIT_CODE       : 0  (0x0)
        SERVICE_EXIT_CODE    : 0  (0x0)
        CHECKPOINT           : 0x0
        WAIT_HINT            : 0x0
```

Additional Discovery Activity

Windows Security Event ID 4799

Windows Security Event ID 4799 recorded enumeration of the local Administrators group.
Observed Process:

```
C:\Windows\SysWOW64\net1.exe
```

Enumerated Group:

```
BUILTIN\Administrators
```

Investigation Analysis

Following successful PsExec execution, the attacker performed group membership discovery to identify privileged local accounts.

This activity is commonly observed during post-exploitation reconnaissance and privilege assessment.

MITRE ATT&CK:

```
T1087 : Account Discovery
```


Security

Number of events: 249 (!) New events available

Keywords	Date and Time	Source	Event ID	Task Category
Audit Suc...	6/1/2026 11:17:13 AM	Microsoft Wi...	4799	Security Grou...

Event 4799, Microsoft Windows security auditing.

General

Details

A security-enabled local group membership was enumerated.

Subject:

Security ID:SYSTEM

Account Name:DESKTOP-HDF09SQ\$

Account Domain:WORKGROUP

Logon ID:0x3E7

Group:

Security ID:BUILTIN\Administrators

Group Name:Administrators

Group Domain:Builtin

Process Information:

Process ID:0x1358

Process Name:C:\Windows\SysWOW64\net1.exe

Log Name:Security

Source:Microsoft Windows security

Logged:6/1/2026 11:17:13 AM

Event ID:4799

Task Category:Security Group Management

Level:Information

Keywords:Audit Success

User:N/A

Computer:DESKTOP-HDF09SQ

OpCode:Info

Session Termination

Windows Security Event ID 4634

Findings

Field	Value
Event ID	4634
User	socadmin
Logon Type	3

Analysis

Windows recorded termination of the remote network logon session.

This event marks the end of the attack lifecycle and correlates directly with the earlier Event

ID 4624.

Security

Number of events: 249

Keywords	Date and Time	Source	Event ID	Task Category
Audit Suc...	6/1/2026 11:17:23 AM	Microsoft Wi...	4634	Logoff

Event 4634, Microsoft Windows security auditing.

General

Details

An account was logged off.

Subject:

Security ID:DESKTOP-HDF09SQ\socadmin

Account Name:socadmin

Account Domain:DESKTOP-HDF09SQ

Logon ID:0x11752BC

Logon Type:

3

This event is generated when a logon session is destroyed. It may be positively correlated with a logon event using the Logon ID value. Logon IDs are only unique between reboots on the same computer.

Log Name:

Security

Source:

Microsoft Windows security

Logged:

6/1/2026 11:17:23 AM

Event ID:

4634

Task Category:

Logoff

Level:

Information

Keywords:

Audit Success

User:

N/A

Computer:

DESKTOP-HDF09SQ

OpCode:

Info

Wazuh Detection Analysis

Rule 92652

Detected successful remote NTLM authentication.

MITRE:

T1078 : Valid Accounts

> Jun 1, 2026 @ 11:16:54.779

input.type: log

agent.ip: 10.0.2.15

agent.name: DESKTOP-HDF09SQ

agent.id: 001

manager.name: wazuh-server

data.win.eventdata.subjectLogonId: 0x0

data.win.eventdata.targetLinkedLogonId: 0x0

data.win.eventdata.impersonationLevel: %%1833

Field	Value
Alert Rule ID	92652
Alert Level	6
Event ID	4624
Event Type	Successful Logon
Username	socadmin
Domain	DESKTOP-HDF09SQ
Source IP Address	192.168.56.107

Field	Value
Source Port	56450
Logon Type	3 (Network Logon)
Authentication Package	NTLM
NTLM Version	NTLM V2
Logon Process	NtLmSsp
Elevated Token	Yes
Computer	DESKTOP-HDF09SQ
Wazuh Description	Successful Remote Logon Detected
MITRE Technique	T1550.002 – Pass the Hash; T1078.002 – Domain Accounts
MITRE Tactic	Initial Access, Lateral Movement, Privilege Escalation, Persistence
Timestamp (UTC)	2026-06-01T05:46:54.779Z

Analyst Observation

A successful NTLM network logon (Event ID 4624) was detected from the Kali Linux system (192.168.56.107) using the administrative account ****socadmin****. The presence of an elevated token and network logon type (3) indicates successful remote authentication, representing the initial access stage of the PsExec remote execution attack chain.

Rule 92218

Detected payload transfer into Windows root directory.
MITRE:

T1570 : Lateral Tool Transfer

> Jun 1, 2026 @ 11:16:54.734	input.type: log	agent.ip: 10.0.2.15	agent.name: DESKTOP-HDF09SQ
	agent.id: 001	manager.name: wazuh-server	
	data.win.eventdata.image: System		
	data.win.eventdata.processGuid: {842f9a06-0195-6a1d-eb03-000000000000}	data.win.eventdata.processId: 4	

Field	Value
Alert Source	Wazuh
Wazuh Rule ID	92218

Field	Value
Rule Level	6
Event Source	Sysmon
Sysmon Event ID	11
Event Type	File Creation
Timestamp (UTC)	2026-06-01 05:46:47.335
Host	DESKTOP-HDF09SQ
User	NT AUTHORITY\SYSTEM
Process	System
Process ID	4
Target File	C:\Windows\ahsjitDb.exe
Detection Description	Possible abuse of Windows admin shares by binary dropped in Windows root folder
MITRE Technique	T1570 - Lateral Tool Transfer
MITRE Tactic	Lateral Movement
Wazuh Alert Group	sysmon, sysmon_eid11_detections, windows

Analyst Observation

Sysmon Event ID 11 recorded the creation of a randomly named executable (C:\Windows\ahsjitDb.exe) in the Windows directory by the SYSTEM process. The file was transferred through the ADMIN\$ share during the PsExec operation. Wazuh correlated the activity as potential Lateral Tool Transfer (MITRE ATT&CK T1570), a common technique used during lateral movement and remote execution attacks.

Rule 92307

Detected registry-based service creation.

MITRE:

T1543.003 : Windows Service

> Jun 1, 2026 @ 11:16:54.796 input.type: log agent.ip: 10.0.2.15 agent.name: DESKTOP-HDF09SQ
agent.id: 001 manager.name: wazuh-server
data.win.eventdata.image: C:\\WINDOWS\\system32\\services.exe
data.win.eventdata.targetObject: HKLM\\System\\CurrentControlSet
\\Services\\Insc\\ImagePath data.win.eventdata.processGuid: {842f

Field	Value
Alert Source	Wazuh
Wazuh Rule ID	92307
Alert Description	Evidence of new service creation found in registry
Severity Level	3
Event Source	Sysmon
Sysmon Event ID	13
Event Type	Registry Value Set
Timestamp (UTC)	2026-06-01 05:46:47
Host	DESKTOP-HDF09SQ
User Context	NT AUTHORITY\\SYSTEM
Process	C:\\WINDOWS\\system32\\services.exe
Process ID	852
Registry Key	HKLM\\System\\CurrentControlSet\\Services\\Insc\\ImagePath
Registry Value	%systemroot%\\ahsjitDb.exe
Service Name	Insc
Payload/Binary	ahsjitDb.exe
MITRE ATT&CK Technique	T1543.003 - Windows Service
ATT&CK Tactic	Persistence, Privilege Escalation

Analyst Observation

Sysmon Event ID 13 recorded a registry modification performed by `services.exe`, creating the service registry path `HKLM\\System\\CurrentControlSet\\Services\\Insc\\ImagePath` and assigning the executable `%systemroot%\\ahsjitDb.exe`. This activity indicates Windows service creation commonly associated with PsExec-style remote execution and lateral movement techniques.

Rule 92650

Severity Level:

12

Detected:

New Windows Service Created to start from windows root path

MITRE:

T1021.002 ; T1569.002

This was the highest-fidelity detection generated during the investigation.

```
> Jun 1, 2026 @ 11:16:54.734 input.type: log agent.ip: 10.0.2.15 agent.name: DESKTOP-HDF09
SQ agent.id: 001 manager.name: wazuh-server
data.win.eventdata.serviceType: user mode service
data.win.eventdata.accountName: LocalSystem
data.win.eventdata.imagePath: %systemroot%\ahsjitDb.exe
```

Field	Value
Alert Rule ID	92650
Alert Level	12 (High Severity)
Alert Description	New Windows Service Created to start from Windows root path
Event ID	7045
Log Source	Service Control Manager
Host	DESKTOP-HDF09SQ
Service Name	Insc
Service File	%systemroot%\ahsjitDb.exe
Service Type	User Mode Service
Start Type	Demand Start
Service Account	LocalSystem
Process ID	852
Event Time (UTC)	2026-06-01 05:46:47
Wazuh Alert Time	2026-06-01 05:46:54
MITRE Technique	T1021.002 – SMB/Windows Admin Shares ; T1569.002 – Service Execution
MITRE Tactic	Lateral Movement ; Execution

Analyst Observation

Wazuh detected the creation of a temporary Windows service (**lnsc**) executing **%systemroot%\ahsjitDb.exe** under the **LocalSystem** account following a remote SMB-based administrative session, indicating PsExec-style remote execution and lateral movement activity.

Event IDs Observed

Windows Security

Event ID	Description
4624	Successful NTLM Logon
4672	Special Privileges Assigned
4634	Session Logoff
4799	Local Group Membership Enumeration

Windows System

Event ID	Description
7045	Service Installed

Sysmon

Event ID	Description
1	Process Creation
11	File Creation
13	Registry Value Set
17	Named Pipe Created
18	Named Pipe Connected

Windows Defender

Event ID	Description
1116	Malware Detection

Event ID	Description
1117	Malware Quarantine

Investigation Findings

The investigation confirmed a complete PsExec remote execution workflow consisting of:

1. Successful NTLM authentication from Kali Linux.
 2. Administrative privilege assignment.
 3. Payload upload through ADMIN\$.
 4. Service installation and execution.
 5. Registry modifications supporting service creation.
 6. Named pipe creation and communication.
 7. Interactive remote shell establishment.
 8. Discovery command execution.
 9. Session termination.
 10. Multiple correlated Wazuh detections.
-

Analyst Assessment

This investigation demonstrates telemetry commonly observed during:

- PsExec abuse
- Impacket activity
- Remote administration abuse
- Lateral movement
- Ransomware deployment
- Post-compromise discovery

The correlated detection chain:

```
4624
↓
4672
↓
Sysmon 11
↓
Sysmon 1
↓
7045
```


↓
Sysmon 13
↓
Sysmon 17
↓
Sysmon 18
↓
Discovery Commands
↓
4634

provides high-confidence evidence of PsExec-based remote execution.

The combination of Windows Security Logs, Sysmon telemetry, Windows Defender detections, and Wazuh alerts enabled full visibility into the attack lifecycle from authentication through remote code execution and post-exploitation activity, making this one of the strongest SOC investigation reports in a portfolio because it demonstrates authentication analysis, execution tracking, service abuse detection, inter-process communication analysis, threat hunting, and SIEM correlation within a single investigation.